



Design patterns running in Super Mario

Scope and how to read

This document describes the **runtime flows that exist in the current source**, not a theoretical list of class relationships. Each section starts from a real situation, then uses a sequence diagram to preserve the exact call order, branches, and object lifetimes. Class, function, and enum names are kept in English so they can be traced back into the code.

Symbol	How to read it in the diagram
->>	Synchronous call; the caller waits until the function/event dispatch finishes.
-->>	Return value or meaningful result (<code>bool</code> , <code>unique_ptr</code> , <code>token</code>).
activate / deactivate	The span during which an object is on the call stack.
alt	An <code>if / switch</code> branch present in the code.
opt	An optional branch that only happens when a runtime condition holds.
loop	A real loop in the code, e.g. bindings or tile codes.

Every arrow in this document is a call/event that can be found in the source; a self-call is used to spell out an internal check or mutation. `EventBus` dispatch is **synchronous**: `notify()` invokes `IObserver::onNotify()` in the same call stack. In contrast, `GameManager` state operations are **deferred** to the safe point at the end of `update()` .

Quick map of the scenarios

Tracked pattern	Runtime scenario	Main seam
Command	One gameplay frame: a held movement key and a pressed fire key	<code>InputHandler</code> -> <code>ICommand</code> -> <code>Mario/Level</code>
Factory Method	<code>Level</code> reads tile <code>g</code> and creates a <code>Goomba</code>	<code>EntityFactory</code> -> <code>EntityCreator</code> -> <code>EnemyCreator</code>
Observer	Coin pickup updates the HUD and plays an SFX	<code>EventBus</code> -> <code>HUD/SoundManager</code>
Game State	<code>Escape</code> opens <code>PauseState</code> via the state stack	<code>IGameState</code> -> <code>GameManager</code>
Mario State	Picking up a Mushroom swaps the power-up state or waits for clearance	<code>Mario</code> -> <code>IMarioState</code>
Singleton (real infrastructure)	The composition root grabs shared managers; <code>Level</code> gets the resource manager	<code>getInstance()</code> -> <code>function-local static</code>

Singleton is traced separately in section 6 because it is lifetime/access infrastructure, not a primary gameplay seam like the five patterns above.

GameManager , EventBus , SoundManager , and TextureManager each have getInstance() /private constructors or the corresponding copy guards.

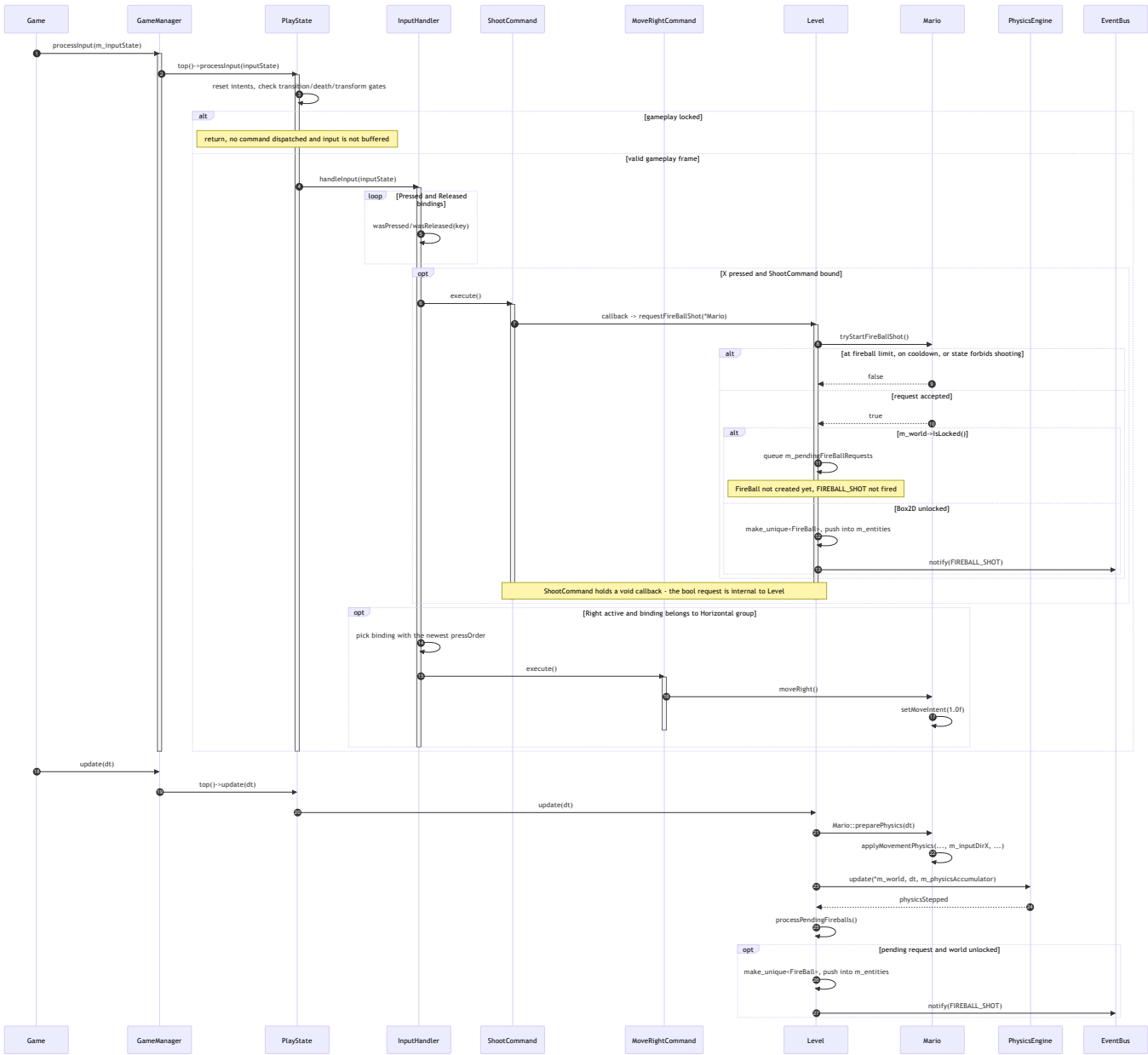
Notably, SaveManager is **not a Singleton**: its constructor is public

(include/core/SaveManager.h:22-26) and GameManager value-owns one object at include/core/GameManager.h:72-75 .

1. Command — turning input into gameplay intent

Scenario: one frame with X pressed and Right held

Game accumulates the InputState from SFML then hands the frame input to the top state. PlayState checks the gameplay gates before passing it to InputHandler . InputHandler knows nothing about Mario or projectiles; it only picks an ICommand::execute() by trigger/group. In the same frame, ShootCommand sends a request to Level while MoveRightCommand sends an intent to Mario .



Roles and where they live in the source

Command role	Actual object	Responsibility
Invoker	InputHandler	Holds <code>unique_ptr<ICommand></code> , checks trigger/group, and calls <code>execute()</code> .
Command	ICommand	The single <code>execute()</code> contract.
Concrete Command	<code>MoveRightCommand</code> , <code>ShootCommand</code> , <code>PauseCommand</code> , <code>RunCommand</code>	Encapsulates one action or callback; does not own <code>Mario</code> / <code>Level</code> .
Receiver	<code>Mario</code> , <code>Level</code>	Receive the intent or request and decide the actual gameplay/physics.

`PlayState::rebindCommands()` installs the bindings (`src/states/PlayState.cpp:62-134`), `Game::update()` and `GameManager::processInput()` push input down (`src/core/Game.cpp:129-130` , `src/core/GameManager.cpp:103-106`). The dispatch loop, the `gameplayEnabled` branch, and the `pressOrder` selection live at `src/patterns/InputHandler.cpp:43-102` . `MoveRightCommand::execute()` calls `Mario::moveRight()` (`src/patterns/MoveRightCommand.cpp:17-21`), while the `ShootCommand` callback goes to `Level::requestFireBallShot()` (`src/states/PlayState.cpp:123-129` , `src/level/Level.cpp:2000-2042`). Inside `Level::update()` , the subsequent physics beat preserves the order `m_mario->preparePhysics(dt) -> PhysicsEngine::update(...) -> processPendingFireballs()` (`src/level/Level.cpp:1294-1317`); the diagram calls `Mario::preparePhysics()` explicitly so it is not mistaken for a non-existent method of `Level` .

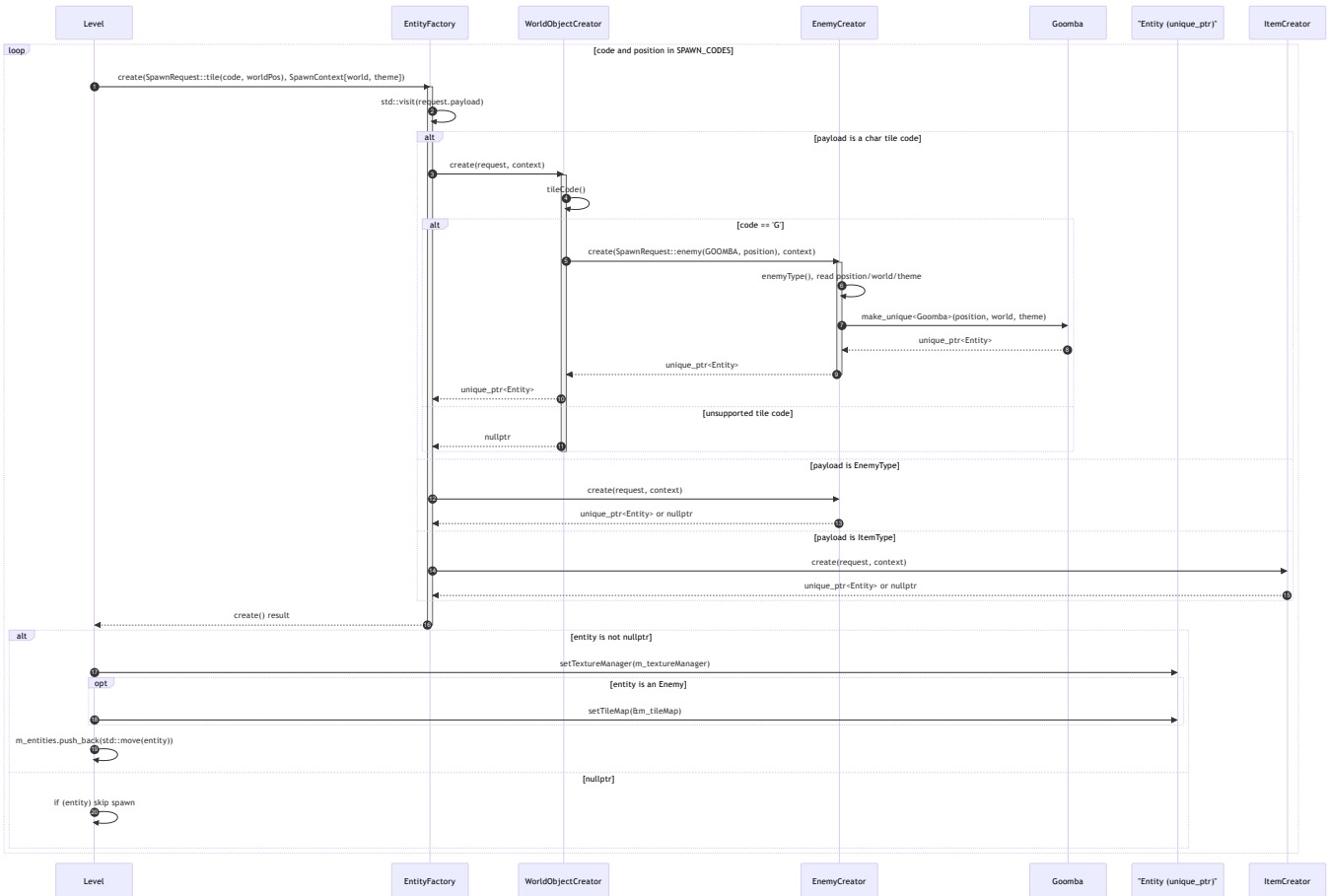
Why the pattern helps

Key mappings can change without touching `Mario` or `Level` ; the same action can also be bound to another keyboard, player 2, or a co-op mode. The `Pressed` , `Held` , `Released` triggers and the horizontal/vertical groups keep the input logic in one place. The trade-off is that commands hold non-owning pointers/callbacks to the receiver; the owner must guarantee `Mario` / `Level` stay alive. `ShootCommand` also only requests a shot: the two-FireBall limit, cooldown, Box2D lock, and entity ownership remain with `Level` .

2. Factory Method — creating entities from a polymorphic request

Scenario: tile G creates a Goomba

`Level` uses the non-static `EntityFactory::create()` seam for the spawn loop. A request carries exactly one payload inside a `std::variant : EnemyType` , `ItemType` , or `char` . For tile `G` , `WorldObjectCreator` converts the tile into an enemy request then delegates to `EnemyCreator` ; only the concrete creator knows which constructor to call.



The `EnemyType / ItemType` branches in the diagram are the shared path for direct callers or compatibility helpers; the production tile-map path mainly goes through the `char` payload and then `WorldObjectCreator` .

Roles and where they live in the source

Factory Method role	Actual object	Evidence
Product	Entity	<code>std::unique_ptr<Entity></code> is the common return type.
Concrete Products	Goomba , Koopa , Mushroom , QuestionBlock , ...	Constructed inside the creators.
Creator seam	<code>EntityCreator::create()</code>	Pure virtual factory method at <code>include/patterns/EntityCreator.h:12-19</code> .
Concrete Creators	<code>EnemyCreator</code> , <code>ItemCreator</code> , <code>WorldObjectCreator</code>	Override the method; <code>WorldObjectCreator</code> also delegates enemy/item creation.
Orchestrator	<code>EntityFactory</code>	<code>std::visit</code> selects the creator at <code>src/patterns/EntityFactory.cpp:19-35</code> .

The real call-site is `Level::spawnEntitiesFromTileMap()` (`src/level/Level.cpp:685-750`). The `G -> Goomba` mapping lives at `src/patterns/WorldObjectCreator.cpp:32-47` , after which `src/patterns/EnemyCreator.cpp:24-78` calls the concrete constructor. The item mapping (`COIN` , `MUSHROOM` , `FIRE_FLOWER` , `STAR`) lives at `src/patterns/ItemCreator.cpp:13-35` .

`EntityFactory::createEnemy()` , `createItem()` , and `createFromTileCode()` are **compatibility static helpers**, not the new canonical seam. They build a `SpawnRequest / SpawnContext` then forward to `defaultFactory().create()` at `src/patterns/EntityFactory.cpp:37-59` ; therefore they must not be interpreted as making `EntityFactory` a Singleton that callers must obtain via `getInstance()` .

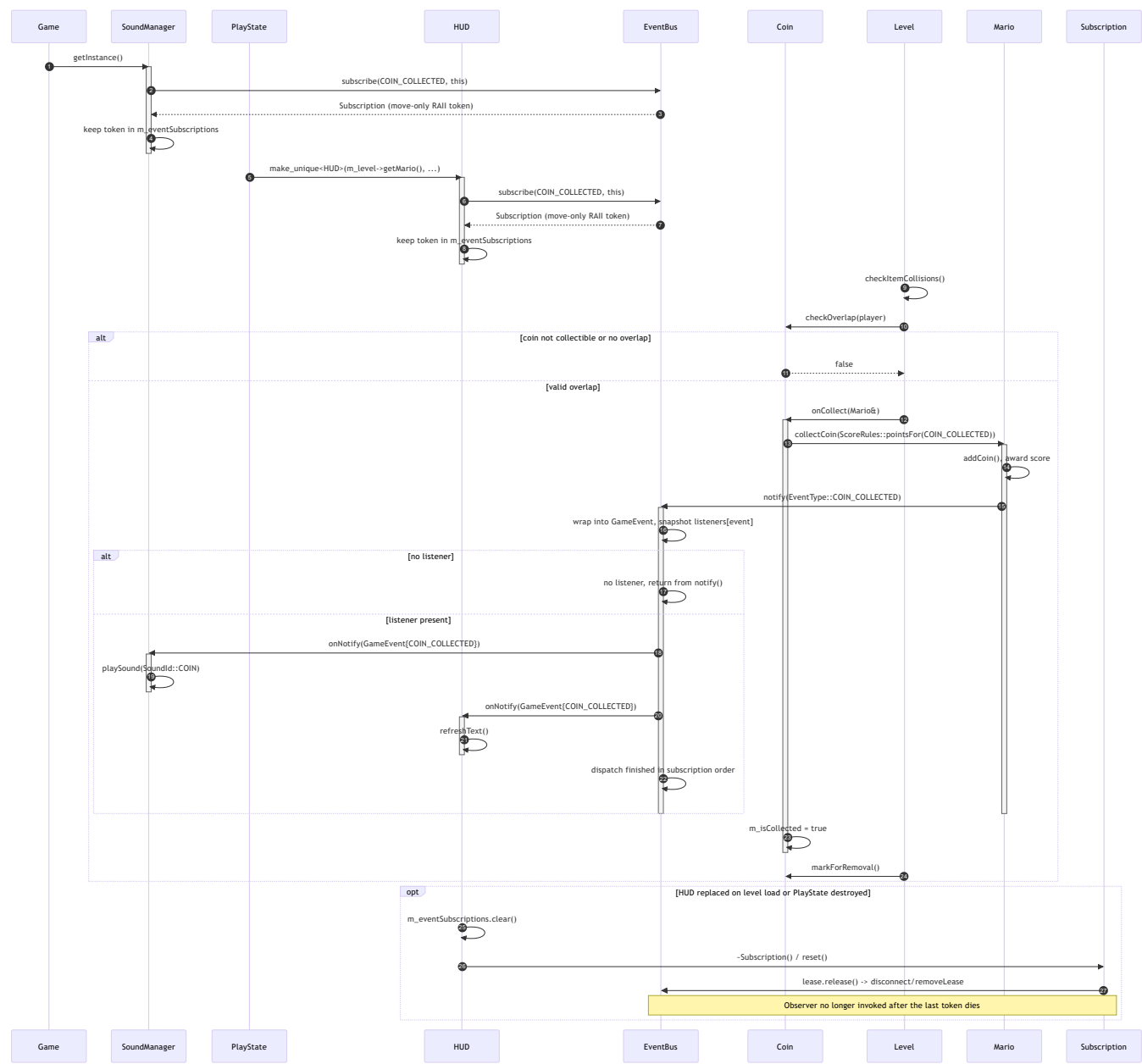
Why the pattern helps

Level knows only the request, the context, and the product base; adding a new enemy/item type can be centralized in a concrete creator instead of sprinkling `new Goomba` , `new Koopa` across the loader. `std::variant` also makes the valid payloads explicit, and creators return `nullptr` for unsupported types/tiles. In exchange, the enum/tile mapping is still a centralized switch; this is a Factory Method with a creator seam, not a promise that every entity self-registers dynamically.

3. Observer — gameplay events reaching the HUD and audio

Scenario: coin pickup, synchronous dispatch to two subscribers

In production, `Game` touches `SoundManager::getInstance()` before gameplay is created, so `SoundManager` registers for events. When `PlayState::loadLevel()` creates the `HUD` , the HUD subscribes to `COIN_COLLECTED` as well. One coin pickup mutates the authoritative data inside `Mario` , then `EventBus` invokes each observer immediately.



Roles and where they live in the source

Observer role	Actual object	Responsibility
Subject	EventBus : ISubject	Stores listeners per EventType , snapshots, and calls onNotify .
Observer	IObserver	The onNotify(const GameEvent&) contract.
Concrete Observers	HUD , SoundManager , PlayState	React independently to the same value event.
Registration lifetime	Subscription	Move-only RAII token; destroying/resetting it disconnects the registration.
Publisher	Mario , Coin , Level , command/collision code	Only publish value events; hold no reference to HUD/audio.

HUD registration and the token lifecycle live at `src/ui/HUD.cpp:112-142` ; audio registers for `COIN_COLLECTED` at `src/core/SoundManager.cpp:45-91` . `HUD::onNotify()` refreshes the display at `src/ui/HUD.cpp:146-190` , while `SoundManager` maps events to SFX at `src/core/SoundManager.cpp:109-184` . The coin pickup flow is invoked from `src/level/Level.cpp:1714-1731` (with a `Box2D` fallback at `src/physics/CollisionManager.cpp:938-949`), then `Coin::awardTo()` -> `Mario::collectCoin()` at `src/items/Coin.cpp:147-168` and `src/entities/Mario.cpp:1392-1402` .

`EventBus::notify()` snapshots then revalidates the lease before each callback (`src/patterns/EventBus.cpp:242-272`), so a callback may reset a subscription without corrupting the loop. If there is no listener, `dispatch` simply returns; this document does not assume a subscriber that the source has not confirmed.

In the diagram, `SoundManager` comes before `HUD` because the composition root calls `SoundManager::getInstance()` before `PlayState::loadLevel()` creates the HUD. This is the registration order of the production path; `EventBus` does not treat observers as running in parallel.

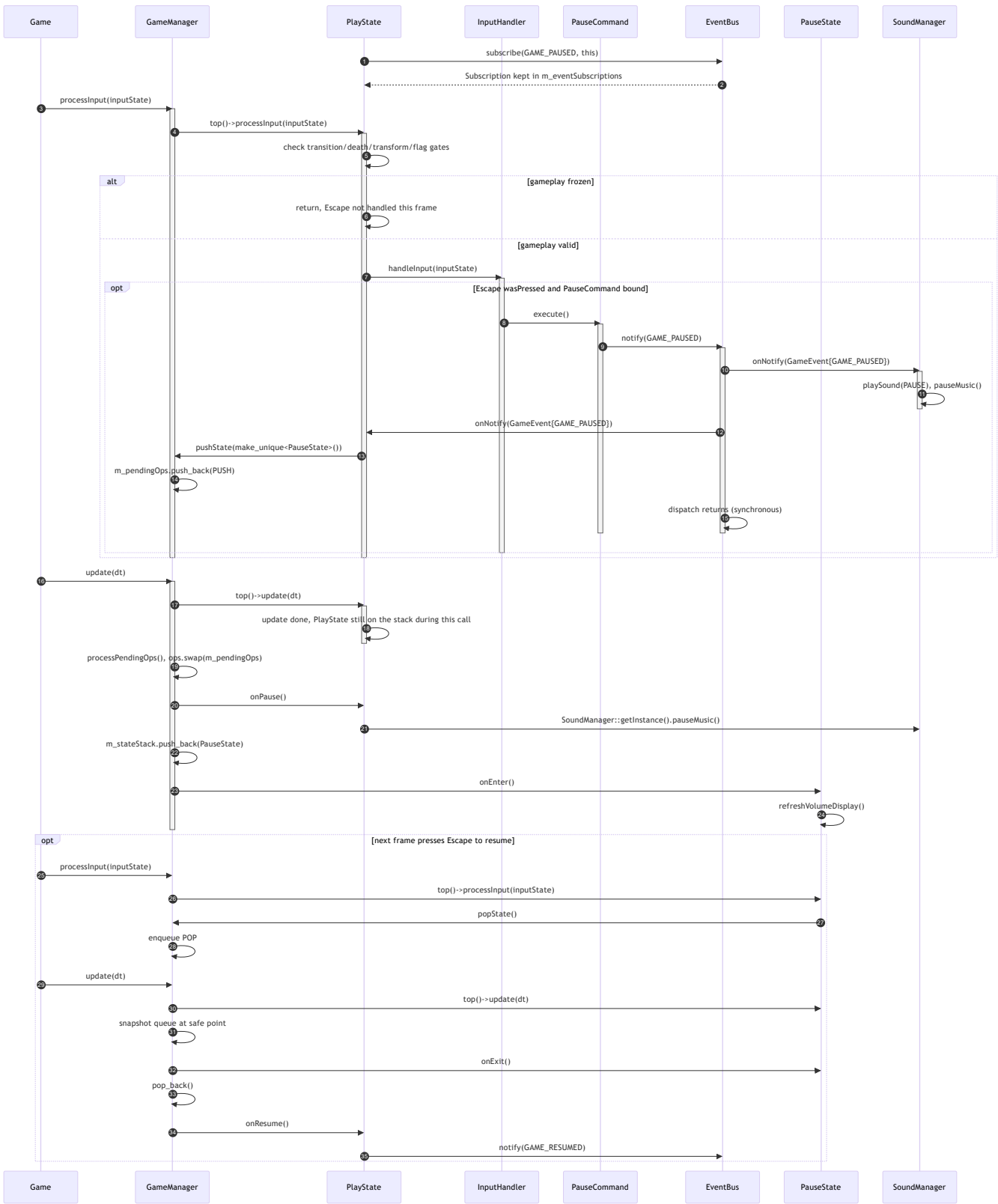
Why the pattern helps

Gameplay code does not need to include or call HUD/SoundManager directly; adding a new observer does not change `Mario::collectCoin()` . The event payload is value-only, so publishers exchange no ownership. The costs are indirection in the control flow and callback ordering that depends on registration order; `dispatch` remains synchronous — it is not a message queue or a background thread. The token must outlive its observer and be kept by a suitable owner.

4. Game State — state stack with deferred transitions

Scenario: Escape opens PauseState at a safe point

`PlayState` does not destroy or replace itself in the middle of input processing. `PauseCommand` publishes `GAME_PAUSED` ; `PlayState::onNotify()` only enqueues a `pushState()` . At the end of `GameManager::update()` , the queue is snapshotted and the new `PauseState` is pushed onto the stack. Because `PauseState::isOverlay()` is `true` , the next render frame can still draw `PlayState` underneath.



Roles and where they live in the source

State role	Actual object	Responsibility
State interface	IGameState	onEnter/onExit/onPause/onResume lifecycle and frame methods.
Concrete states	MenuState , PlayState , PauseState , GameOverState , WinState , ...	Encapsulate the behavior of each mode.
Context/owner	GameManager	Forwards event/input/update to the top state and owns the stack.
Transition policy	PendingOp { CHANGE, PUSH, POP }	Separates the state-change request from the moment objects are destroyed.

GameManager::changeState/pushState/popState() only append a pending operation (src/core/GameManager.cpp:32-42). update() calls the top state first and only then processPendingOps() (src/core/GameManager.cpp:76-95); applyOp(PUSH) calls onPause , pushes the object, and calls onEnter (src/core/GameManager.cpp:44-73). PlayState registers GAME_PAUSED /enqueues

the push at `src/states/PlayState.cpp:281-296` and
`src/states/PlayState.cpp:349-410`; `PauseState` resumes via `popState()` at
`src/states/PauseState.cpp:287-291`.

`Bus -> SoundManager -> PlayState` in the diagram reflects the production registration order: `SoundManager` receives `GAME_PAUSED` first, then `PlayState` enqueues the `PUSH`. `PlayState::onPause()` also calls `pauseMusic()` again when the operation is applied; both calls are real and must not be collapsed into one synchronous transition.

The queue is snapshotted via `ops.swap(m_pendingOps)`. Consequently, if a lifecycle callback creates additional operations, the new operations land in the now-empty queue and wait for the next safe point; this must not be described as an instantaneous transition inside the current call stack.

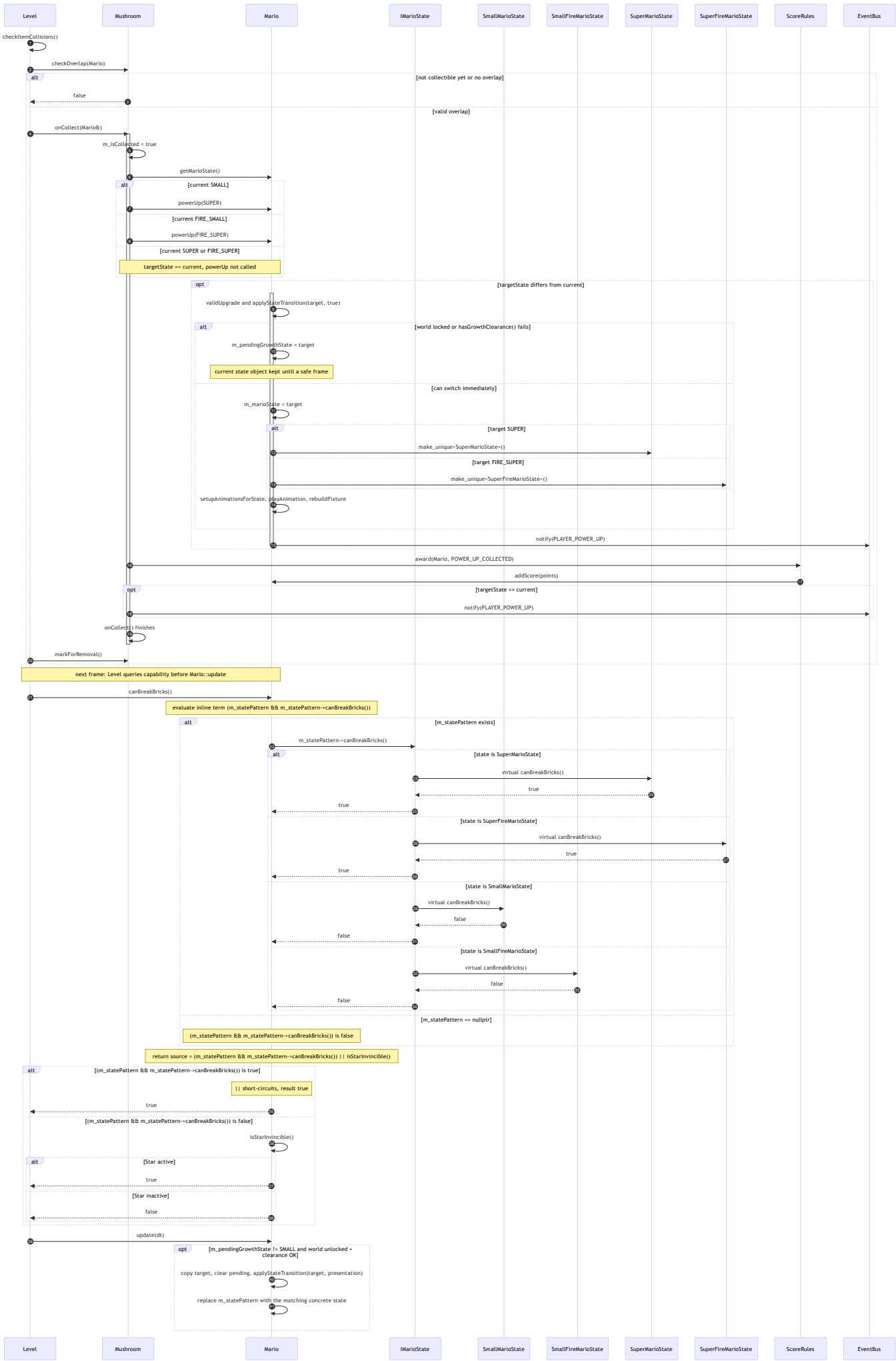
Why the pattern helps

`GameManager` needs no giant `switch` for menu/play/pause/game-over; each state owns its behavior and lifecycle. The stack allows a pause overlay while keeping the play state underneath. The trade-off is understanding `CHANGE` (clears the whole stack), `PUSH` (overlay), and `POP` (resumes the state below); every state operation is delayed at least until the end of `update()`.

5. Mario State — power-up states change capabilities

Scenario: picking up a Mushroom, either growing immediately or waiting for clearance

`Mushroom::onCollect()` reads the current `MarioState` and picks the target state. For `SMALL -> SUPER` OR `FIRE_SMALL -> FIRE_SUPER`,
`Mario::powerUp()` swaps `m_statePattern` for a concrete `IMarioState`. If `Box2D` is locked or there is not enough headroom, `applyStateTransition()` keeps `m_pendingGrowthState`; the pickup event is still published, but the state object does not change in that frame. This is Mario's own deferred growth, not the `GameManager` state queue.



Roles and where they live in the source

State role	Actual object	Responsibility
Context	Mario	Holds <code>m_marioState</code> and the <code>unique_ptr<IMarioState></code> .
State interface	<code>IMarioState</code>	Interface declaring capabilities and lifecycle hooks; production currently only delegates <code>canBreakBricks()</code> through the state pointer.
Concrete states	<code>SmallMarioState</code> , <code>SuperMarioState</code> , <code>SmallFireMarioState</code> , <code>SuperFireMarioState</code>	Return different capabilities and represent power tiers.
Transition policy	<code>Mario::applyStateTransition()</code>	Checks body/clearance, builds the state object, fixture, and presentation.

The item pickup path lives at `src/level/Level.cpp:1714-1731` ; the target state and re-entry branches live at `src/items/Mushroom.cpp:106-140` . Building the concrete state, growth deferral, and fixture rebuild live at `src/entities/Mario.cpp:963-1074` ; the next-frame loop flushes pending growth at `src/entities/Mario.cpp:465-475` . Level asks for the capability before handling the tile hit at `src/level/Level.cpp:1325-1329` , while `Mario::canBreakBricks()` delegates to `m_statePattern` at `src/entities/Mario.cpp:1490-1493` . In contrast, `Mario::canShootFireBall()` does not call `IMarioState::canShootFireBall()` ; it checks `usesFire(m_marioState)` and `m_fireCooldown <= 0.0f` directly (`src/entities/Mario.cpp:1486-1488`).

`IMarioState` declares `onEnter` , `onExit` , and `update` in the interface (`include/states/IMarioState.h:17-33`), but the current transitions **do not invoke those callbacks directly**; the source only creates/replaces the `unique_ptr` , updates animation/fixture, and uses the virtual capabilities. For the same reason, production has no call-sites for `IMarioState::getHitboxSize()` , `canShootFireBall()` , `onEnter()` , `onExit()` , or `update()` ; these are declared hooks/contracts, not runtime delegation being traced here. The diagrams therefore do not invent those calls.

Re-entry and damage are preserved correctly

A Mushroom picked up while Mario is already `SUPER` / `FIRE_SUPER` does not call `powerUp` ; it still awards points and publishes exactly one `PLAYER_POWER_UP` (`src/items/Mushroom.cpp:127-139`). FireFlower follows the same principle for `SMALL/SUPER` and the fire tier (`src/items/FireFlower.cpp:51-75`). For damage, `Mario::powerDown()` is ignored while immune/star-powered/dying/transforming; the valid transitions are `FIRE_SUPER -> SUPER` , `FIRE_SMALL -> SMALL` , `SUPER -> SMALL` , while `SMALL` calls `loseLife()` (`src/entities/Mario.cpp:1137-1161`). This is why not every collision should be interpreted as a state change.

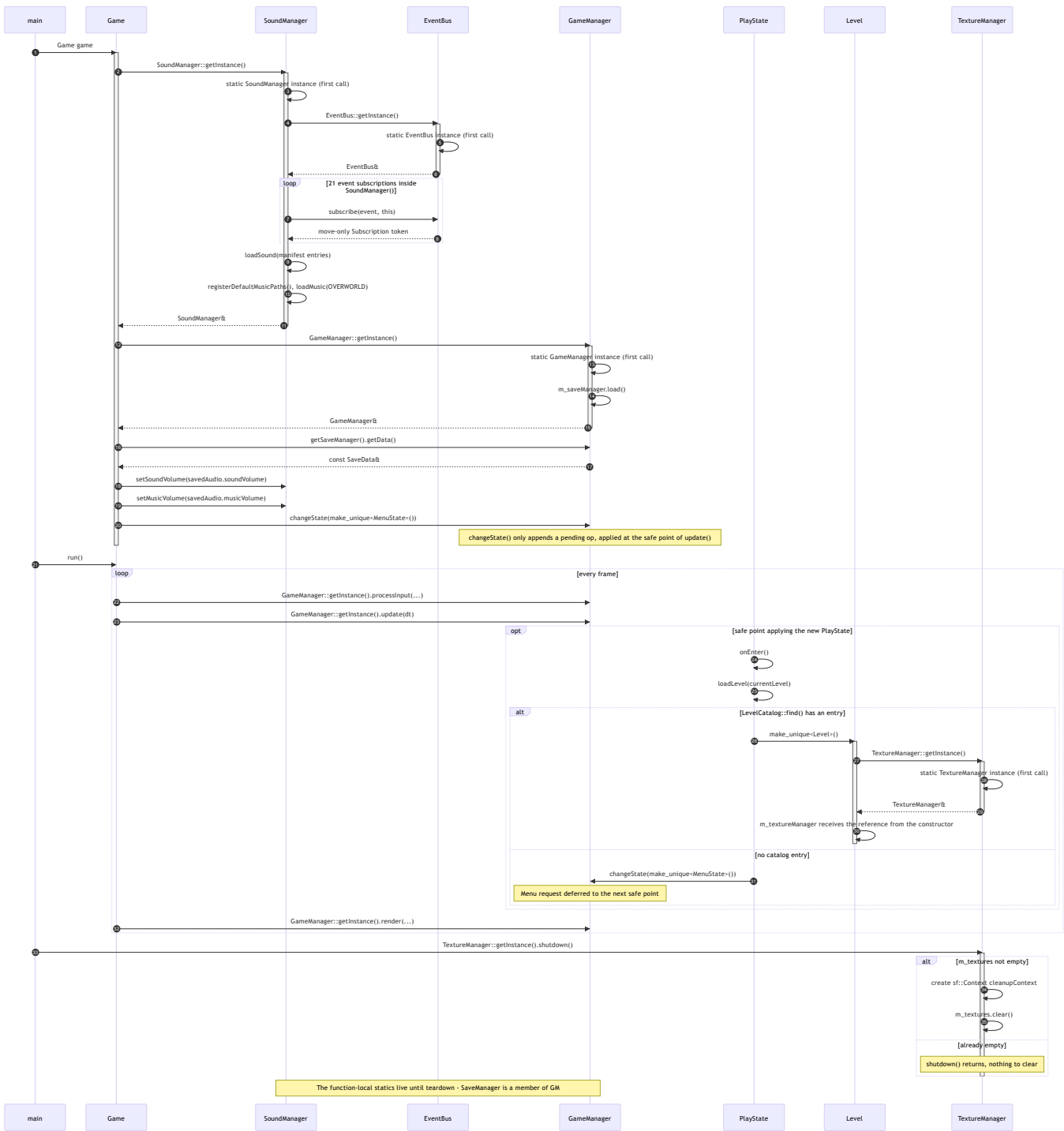
Why the pattern helps

Gameplay code delegates the `canBreakBricks()` capability instead of scattering per-tier conditionals; the FireBall firing condition currently remains an enum/cooldown check inside `Mario` . Swapping the state object allows new tiers without changing callers. The trade-off is that body/fixture and animation must stay in sync with the state; blocked growth needs a pending marker so Box2D is not mutated while the world is locked. This state is also not an async state machine: swapping the concrete object is usually synchronous — only growth is held back until a safe `Mario::update()` .

6. Singleton — shared infrastructure with a bounded lifetime

Scenario: the composition root grabs managers, then `Level` gets the resource manager

This is a genuine Singleton scenario, kept apart from the gameplay seams in sections 1-5. `Game::Game()` calls `SoundManager::getInstance()` first so its constructor registers observers and preloads assets, then obtains `GameManager::getInstance()` to read `SaveManager` and queue `MenuState` . When `PlayState::loadLevel()` creates a `Level` , the `Level` constructor grabs the same `TextureManager` reference. At process end, `main()` calls `TextureManager::shutdown()` while the graphics context is still alive.



Roles and where they live in the source

Singleton role	Actual object	Responsibility and evidence
Instance/accessor	GameManager , SoundManager , TextureManager , EventBus	getInstance() returns a function-local static at src/core/GameManager.cpp:27-30 , src/core/SoundManager.cpp:40-43 , src/core/TextureManager.cpp:15-18 , src/patterns/EventBus.cpp:196-208 .
Construction/copy guard	The four participants above	GameManager has a private constructor/destructor and deleted copies (include/core/GameManager.h:23-50); SoundManager has the accessor, deleted copy/move, and private constructor/destructor (include/core/SoundManager.h:116-125 , include/core/SoundManager.h:211-214); TextureManager has deleted copies and a private constructor/destructor (include/core/TextureManager.h:25-35 , include/core/TextureManager.h:70-80); EventBus has a private constructor/destructor and deleted copies (include/patterns/EventBus.h:36-55).
Instance clients	Game , PlayState , Level , main	Game gets the Sound/Game managers and the first state (src/core/Game.cpp:62-76); PlayState::onEnter() calls loadLevel() (src/states/PlayState.cpp:281-304) then creates Level (src/states/PlayState.cpp:583-601); Level takes the texture reference (src/level/Level.cpp:174-175); main calls shutdown (src/main.cpp:10-16).
Lifetime/resource boundary	SoundManager , TextureManager	The Sound destructor clears subscription tokens (src/core/SoundManager.cpp:105-107); TextureManager::shutdown() clears GPU resources under an sf::Context (src/core/TextureManager.cpp:123-132).

The flow above shows the concrete benefit: states/entities do not have to thread a global registry through every constructor, and Level keeps one reference to the single resource manager. The trade-off is that a global accessor hides dependencies and requires test/lifetime discipline; this is why GPU cleanup is invoked explicitly in main() . Do not infer from this that SaveManager is a Singleton: its constructor is public (include/core/SaveManager.h:22-26), and GameManager value-owns the m_saveManager member (include/core/GameManager.h:69-75).

Places not to mislabel as patterns

- 1. SaveManager is not a Singleton. Its public constructor allows independent instances in tests/sessions; the production composition root

lets `GameManager` value-own one instance accessed through
`getSaveManager()` .

2. `EntityFactory::createEnemy/createItem/createFromTileCode` are compatibility forwarding helpers. `defaultFactory()` uses a static local to avoid duplicating the mapping, but the canonical API remains `EntityFactory::create(request, context)` and `EntityFactory` has a public constructor.
3. `GameManager` , `EventBus` , `SoundManager` , and `TextureManager` have real singleton accessors; section 6 traces this lifetime/access behavior, but singleton storage does not replace the Observer/Command/State seams described here.
4. The arrows in the diagrams are not class-diagram relations: only calls/events with live call-sites are drawn; subscribers, return values, and ownership are annotated wherever the source provides evidence.